— THE RATTY2TAILS PAPERS —

The Ledger and the Stone

Saleability, time preference, and the long search for a money nobody can print

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Monetary History Series

Contents

EDITOR'S NOTE

This essay is an original synthesis, not a summary of any single book. It draws on a tradition running from Carl Menger through Ludwig von Mises and Friedrich Hayek to contemporary popularisers such as Saifedean Ammous, whose *The Bitcoin Standard* did much to bring this way of thinking about money to a wider audience. Where that tradition's claims are contested by mainstream economists — and several of them are, particularly on business cycles and on the cultural effects of monetary regimes — I've tried to say so rather than present one school's conclusions as settled fact. Treat this as a guided tour of an argument, not a verdict on it.

CHAPTER ONE

The Original Problem

Every economy above a certain size runs into the same wall. A woman who bakes bread wants a new roof; the roofer wants a violin lesson for his daughter; the violin teacher wants bread. Nobody in this little triangle can trade directly with the person who has what they want, and as the circle of strangers grows, from a village to a city to a continent, the odds of any two people's wants lining up neatly fall towards zero. Economists call this the problem of the coincidence of wants, and it is not a minor inconvenience. It is the wall that direct exchange, or barter, runs into as soon as a society tries to do anything more complicated than subsist.

The trouble is not merely that the baker and the roofer want different things. It runs deeper, into three distinct dimensions that any workable medium of exchange has to solve at once. The first is scale: a roof is worth vastly more than a loaf of bread, and bread does not divide into fractions of a roof in any sensible way. The second is time: bread goes stale in days, but the roofer wants to be paid across the months it takes to shingle a house, and the baker cannot very well hand over a warehouse of bread today against a promise that will only be fulfilled at the end of the season. The third is space: the wool merchant three valleys over might have exactly what the shepherd wants, but neither party can be bothered to haul cartloads of raw wool over a mountain pass on the chance that the other side's preferences have not changed since the last message arrived.

What breaks the deadlock is the emergence of an intermediate good: something accepted not because anyone particularly wants to consume it, but because everyone expects everyone else to accept it in turn. Once such a good becomes widely enough recognised, the baker no longer needs to find a roofer who wants bread. She only needs to find somebody, anybody, willing to buy bread for the intermediate good, and then use that good to buy roofing. This is the origin of money, and it is worth pausing on how strange the arrangement actually is. Money is the one economic good that is not wanted for its own sake and not wanted as an input into production. It is wanted purely because other people want it too — a self-reinforcing social agreement dressed up as an object.

Why this problem never fully goes away

It would be convenient to treat the emergence of money as a historical event that, once accomplished, settled the matter permanently. It did not. Every generation re-discovers some version of the coincidence-of-wants problem, usually at the exact moment its existing monetary arrangements come under strain: during a currency collapse, in a besieged city cut off from its usual mint, in a prison camp where the official currency is confiscated at the gate. In each of these settings, something unlikely and almost always unplanned steps into the breach — tinned mackerel, phone credit, cigarettes — and a miniature monetary system reconstitutes itself within days. The persistence of this pattern is itself a kind of evidence: whatever money is, it is not merely a convention stamped onto paper by a sovereign. It is a solution that markets keep re-inventing whenever the usual solution is taken away.

This matters for what follows, because the question this essay is really asking is not "what is money" in the abstract, but a narrower and more useful one: given that people will always need *some* good to serve this function, which candidates tend to do the job well, which tend to do it badly, and what happens to the societies that end up choosing badly, whether through misfortune or through the deliberate choice of whoever controls the mint. That is a question about the properties of goods, not about decrees, and it is where the Austrian economist Carl Menger begins.

CHAPTER TWO

Menger's Test

In 1892, Carl Menger published a short essay, "On the Origin of Money," that answered a question which had puzzled economists for a century: why do societies converge on one or two goods as money, out of the enormous range of things they could in principle use? Menger's answer was that money is not decreed into existence by a ruler or a council. It emerges spontaneously, the unplanned product of countless individuals each trying to make their own trades easier, in the same way that footpaths emerge across a park without any planning committee deciding where they should go.

The property that decides which goods win this race, Menger argued, is **saleability**: how easily a good can be sold, at short notice, without a significant loss in its price. Some goods are highly saleable — a stranger arriving in almost any town can convert them to local currency within the hour. Others, however useful in their own right, are saleable only slowly and at a discount, because finding a buyer takes time, or because the buyer who does show up knows the seller is desperate and prices accordingly. It is this property, and not any decree, that pushes an economy toward using one or two goods as money rather than fifty.

Three axes of saleability

It is useful to break saleability down into three separate questions, because a good can score well on one and badly on another, and the history of money is largely a history of goods that solved two of the three and were eventually displaced by something that solved all three at once.

Dimension	The question it answers	A good that struggles here
Across scale	Can it be divided or combined to match any transaction size, from a loaf of bread to a house?	Cattle — you cannot sell a fifth of a cow and expect it to still be a cow
Across space	Does it carry a lot of value in a little bulk, so it can travel with a trader or a refugee?	Salt or grain — valuable, but bulky relative to their worth
Across time		

	Will it still be wanted, and wanted at roughly the same price, next year or next decade?	Fresh fish — perfectly saleable today, worthless by Thursday

The third of these, saleability across time, turns out to be the decisive one, and it is where this essay will spend most of its attention. A good can be perfectly divisible and perfectly portable and still fail as money if its supply can be casually increased whenever people start wanting to hold it as savings. This is not a minor technical footnote. It is, on the view this essay is exploring, the central fact that explains almost the entire monetary history of the human species — and it deserves its own chapter.

> *Menger's insight, stripped of jargon, is that money is discovered, not decided. Whatever a society eventually uses as money is simply whatever won an unplanned contest for saleability — which is also why attempts to legislate a new money into existence so often fail, and why attempts to legislate an existing one out of existence so often fail too.*

CHAPTER THREE

Stock, Flow, and the Arithmetic of Debasement

Here is a thought experiment. Imagine a small island economy that decides, for want of anything better, to use a particular seashell as money. The shells are pretty, portable, and hard to find on the local beaches, so for a generation they work rather well. Then a trading ship arrives with a hold full of identical shells, gathered easily from a reef two hundred miles away, and offers to trade them for the island's yams, timber, and labour. What happens next is not really a story about seashells. It is a story about arithmetic, and it applies with only the details changed to Roman coinage, Weimar banknotes, and, its defenders would argue, to gold and Bitcoin as well.

The ratio that matters

The relevant arithmetic is the ratio between a good's existing stockpile — everything ever produced and not yet destroyed or consumed — and its flow, the new quantity that will be produced in the coming year. Call this the stock-to-flow ratio. A good with a high ratio has an existing stockpile so much larger than any plausible new production that even a dramatic surge in mining, minting, or harvesting barely moves the needle on total supply. A good with a low ratio can have its supply meaningfully inflated in a single season, because the existing stockpile was never that large to begin with.

This ratio matters because of what economists sometimes call the store-of-value trap, and it works the same way regardless of which good falls into it. The moment any good starts being held as savings rather than simply consumed, demand for it rises above its ordinary consumption demand, and its price rises with it. If the good has a low stock-to-flow ratio, that price rise is a signal to producers: make more, and make it now, while the price is high. If producers can respond quickly — and with anything that is not geologically or chemically constrained, they usually can — new supply floods in, the price collapses, and the people who trusted the good as a store of value are left holding a devalued asset while the producers who flooded the market walk away with the gains. The wealth has not vanished; it has been transferred, quite mechanically, from savers to producers.

A good with a genuinely high stock-to-flow ratio is comparatively immune to this trap, not because of any virtue on the part of its holders, but for the boring reason that no plausible increase in annual production is large relative to the mountain of existing stock. This is arithmetic, not ideology, and it is worth being precise about it before drawing any larger conclusions from it.

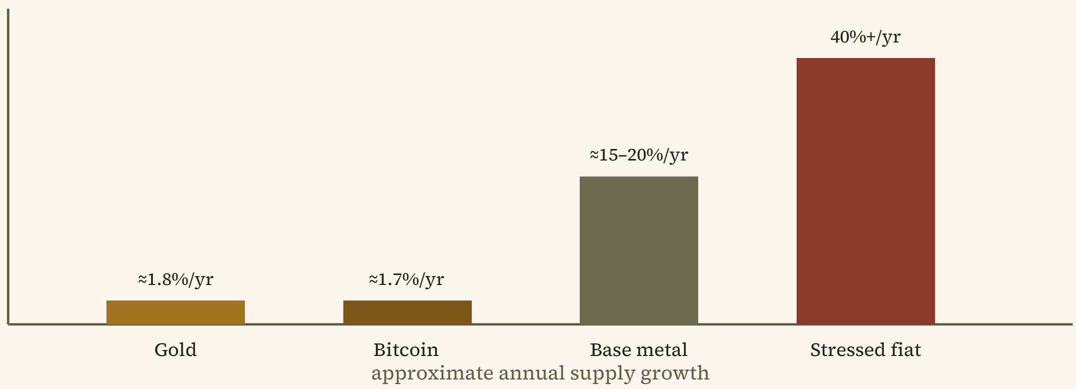

Figure 1 — Illustrative, not precise: approximate annual supply growth for four categories of monetary good. The exact figures move year to year and are contested at the margins; the ordering is the point.

A necessary caveat

None of this means a high stock-to-flow ratio is sufficient, on its own, to make a good functional as money. Diamonds have a reasonably high ratio and have never seriously functioned as everyday money, partly because their value depends on subjective grading that ordinary people cannot easily verify. The ratio is a necessary filter, not a complete theory. It explains why certain candidates are disqualified quickly; it does not, by itself, explain everything about why the survivors succeed. Keeping that distinction in view will matter later, when this essay turns to Bitcoin, because stock-to-flow arguments are frequently presented as though they settle questions — about volatility, about adoption, about long-run value — that they do not actually settle.

CHAPTER FOUR

A Cabinet of Curiosities

The historical record is, frankly, a bit of a museum of the strange things human beings have used as money, and each exhibit teaches the stock-to-flow lesson from a slightly different angle. None of what follows is exhaustive; it is a small selection chosen because each case illustrates a different way a monetary good can rise, and a different way it can fall.

The stone money of Yap

On the Micronesian island of Yap, communities for centuries used enormous carved limestone discs, some weighing several tonnes, as a form of high-value money. What makes the Yapese system genuinely remarkable is that the stones frequently never moved at all after being installed in a village; ownership simply changed hands by common agreement and communal memory, in effect a public ledger maintained by social consensus rather than by any physical transfer. The stones worked as a store of value for a simple reason: limestone does not occur naturally on Yap, and quarrying a stone meant sailing to Palau, cutting it by hand, and rafting it hundreds of kilometres home — an effort so large that the existing stock of stones dwarfed anything a single expedition could add to it. The arrival of steel tools and motorised boats in the nineteenth century changed that calculation permanently. Once stones could be quarried and shipped with a fraction of the earlier effort, the supply became elastic to demand in a way it never had been before, and the stones drifted from currency into ceremonial heirloom, a role they still occupy on Yap today.

Cowrie shells and the double-edged sword of trade

Cowrie shells served as money across large stretches of Africa, South Asia, and Oceania for a very long stretch of human history, prized because they were durable, difficult to counterfeit, and, in most of the regions that used them, genuinely hard to obtain locally. Their undoing illustrates a variant of the same lesson as the Yapese stones, but driven by trade rather than technology: European trading companies, with access to Indian Ocean cowrie beds unfamiliar to inland African economies, began importing shells in bulk during the height of the Atlantic trade. The resulting glut debased local cowrie currencies over the space of a few generations, with consequences that were not merely financial —

historians studying West African economic history have connected the devaluation of cowrie money to the wider and darker economic disruptions of the era. It is a sobering illustration that a monetary collapse driven by an oversupply of the underlying good is not an abstract inconvenience; it can hollow out the savings of an entire region while the people supplying the glut profit handsomely.

Tobacco, cigarettes, and the economics of a prison camp

A gentler but no less instructive case comes from the economist R. A. Radford's 1945 account of life in a Second World War prisoner-of-war camp, where the official currencies of the prisoners' home countries were useless behind the wire and cigarettes spontaneously took over as a functioning currency, complete with prices, informal banking, and a recognisable business cycle driven by the arrival and depletion of Red Cross parcels. Cigarettes worked precisely because their supply, inside the camp, was bottlenecked by parcel deliveries rather than by anything the prisoners themselves could increase at will. It is a small, almost charming illustration of the same underlying mechanism as the Yapese stones and the West African cowries, playing out in miniature and over months rather than centuries.

Cattle, salt, and the words we still use

Cattle and salt both served monetary functions across large parts of the ancient world, and both illustrate the opposite failure mode from the previous examples: rather than being undone by a sudden glut, they were simply outcompeted on saleability across scale. A cow cannot be divided into a hundred equal instalments without ceasing to be a cow, which made cattle serviceable for large transactions, such as dowries, but hopeless for buying bread. Salt solved the divisibility problem but was heavy relative to its value and travelled poorly. Both left their mark on language rather than on any surviving currency: the Latin word for cattle, *pecus*, is the root of "pecuniary," and the Latin word for salt, *sal*, sits behind "salary" — a linguistic fossil record of monetary systems that metal coinage eventually displaced.

A PATTERN WORTH NAMING

Notice that none of these moneys failed because people stopped wanting them, and none failed because a government banned them. Each failed, in one way or another, when the cost of producing more of the good fell — through new tools, new trade routes, or new sources of supply — faster than the good's holders could adjust their expectations. The lesson generalises: a monetary good's soundness is not a fixed property of the object. It is a running bet on how hard the good will remain to produce, and that bet can be lost by any monetary good, including the metals that came next.

CHAPTER FIVE

Gold's Long Reign

Metals solved the divisibility problem that cattle could not and the portability problem that salt could not, and among the metals, gold eventually won out over silver, copper, and bronze for reasons that are almost entirely chemical rather than cultural. Gold does not tarnish, corrode, or react with virtually anything it is likely to encounter, which means that, unlike iron or even silver, essentially all the gold ever mined by humanity still exists somewhere — in vaults, in jewellery, in shipwrecks waiting to be found. And gold cannot be synthesised from cheaper materials; centuries of alchemists tried and failed for the excellent reason that gold's identity as an element is fixed by its atomic structure, not by anything a furnace can rearrange.

These two facts together produce an extraordinarily high stock-to-flow ratio, historically in the range of one to two percent of annual growth against the accumulated stockpile of millennia, and that ratio is the real explanation for gold's monetary dominance, not any inherent shininess. Even a genuine gold rush, discovering a rich new seam and doubling the annual mining rate, moves the needle on total above-ground gold by only a percentage point or two, because the existing stockpile is the accumulated product of thousands of years of mining. No other widely available material comes close to this combination of chemical stability and mining difficulty, which is why silver, useful as it was for smaller everyday transactions, was gradually squeezed out of monetary use once banking technology made it possible to settle large payments in gold-backed paper instead.

The trap gold could not avoid

Gold's very success, however, created a problem that had nothing to do with its chemistry. As trade expanded across the nineteenth century, physically transporting gold coins for every transaction became impractical, and banks stepped in with paper receipts, cheques, and eventually central bank notes, each nominally redeemable for a fixed weight of gold sitting in a vault. This was a genuine improvement in convenience, and for several decades, roughly from 1870 to 1914, it underwrote a period of remarkably open global trade and rapid capital accumulation that economic historians still study as something of a high-water mark.

But the arrangement required centralising the actual metal in a small number of vaults, and centralisation of anything valuable eventually attracts the attention of whoever controls the state apparatus nearby. Banks, and then central banks, discovered they could issue more paper claims than they held gold to redeem, so long as depositors did not all ask for their gold on the same afternoon. This is not a conspiracy theory; it is a description of how fractional-reserve banking works, and it is precisely the vulnerability that the First World War exposed when the major combatants suspended gold convertibility within weeks of the fighting starting, freeing themselves to print money to finance the war in a way that direct taxation of a wary public would never have permitted so quickly. Gold's chemistry made it nearly impossible to counterfeit or dilute directly; it did nothing to stop governments from diluting the paper claims layered on top of it. That gap between the metal and the certificate is the single most important fact in twentieth-century monetary history, and it deserves the next chapter to itself.

> *It is worth sitting with the irony here: the very feature that made gold excellent money — its need for physical, centralised storage once transaction volumes grew large — is also the feature that made it capturable by the one kind of institution with both the incentive and the legal authority to dilute a currency at scale.*

CHAPTER SIX

The Fiat Century

The First World War did not merely interrupt the gold standard; it inaugurated a century in which the link between money and any physically constrained good would be loosened, tightened, renegotiated, and finally cut altogether, in stages that are worth walking through because each stage solved a real problem while creating a new one.

Bretton Woods and its slow unwinding

The 1944 Bretton Woods conference attempted a compromise: the US dollar would be pegged to gold at a fixed rate, and other major currencies would be pegged to the dollar, giving the postwar world something resembling a gold standard at one remove. For roughly two decades this arrangement coexisted, not entirely comfortably, with the reality that the United States was financing a growing menu of domestic and international commitments partly through money creation. Foreign governments, watching the gap between the quantity of dollars in circulation and the quantity of gold in Fort Knox widen, began converting their dollar reserves back into gold at an accelerating pace during the 1960s. By August 1971, with gold reserves under mounting pressure, President Richard Nixon suspended dollar-gold convertibility entirely, an act intended as temporary that has, more than fifty years later, never been reversed. From that point forward, essentially every major currency on Earth has been fiat: valuable because governments require taxes to be paid in it and because habit and network effects keep people using it, but backed by no external constraint on its supply beyond the discretion of the institutions that issue it.

What this bought, and what it cost

It is worth being fair to the fiat era before criticising it, because the criticism only has force if the trade-off is stated honestly. Freed from a gold peg, central banks gained the ability to expand the money supply rapidly during acute crises — the 2008 financial crisis and the 2020 pandemic shutdown both saw extraordinary monetary interventions that many mainstream economists credit with averting a considerably deeper depression than either event might otherwise have produced. Milton Friedman, no friend of central bank overreach, spent much of his career arguing that the Federal Reserve's passive tolerance of a shrinking money supply, not any excess of activity, was the chief cause of the Great Depression's severity — a view that sits uneasily with the harder-line Austrian argument that monetary authorities should not be actively managing the money supply at all. This is a genuine and unresolved dispute within economics, not a case where one side has been quietly debunked, and readers should treat any confident-sounding resolution of it, in either direction, with a healthy degree of suspicion.

What is much less disputed is the raw arithmetic of what a fiat regime does to the purchasing power of a currency held over long periods, absent an anchor. The US dollar has lost roughly ninety-five percent of its purchasing power against a basket of ordinary consumer goods since 1971, a fact easily verified against published inflation indices, and hyperinflations — Weimar Germany, Zimbabwe, and, at the time of writing, Venezuela and Argentina at various points over the past decade — remain a phenomenon almost exclusively associated with fiat currencies rather than commodity-backed ones. Researchers who study the history of hyperinflation count several dozen episodes meeting a strict statistical definition since the early twentieth century, and all but a small handful occurred under government-issued paper money untethered from any external constraint.

A NECESSARY DISTINCTION

There is an important difference between "fiat currencies are capable of severe mismanagement" and "fiat currencies are necessarily mismanaged." The currencies of Switzerland, Singapore, and, for long stretches, the United States and the United Kingdom have held their value tolerably well by historical standards even without a metal peg. The stronger Austrian claim — that central management of a currency's supply is inherently and inevitably corrosive over sufficiently long horizons — is a real position in the economics literature, but it remains a contested one, not a demonstrated law of nature.

Monetary nationalism and the cost of floating exchange rates

One underappreciated consequence of the move to fiat currencies, flagged by Friedrich Hayek in his 1937 essay on what he called monetary nationalism, is the sheer administrative weight now carried by the global foreign exchange market: trillions of dollars change hands daily simply to convert one national currency into another, a cost of doing business across borders that a shared gold standard, whatever its other flaws, never required. Whether this cost is a price worth paying for the flexibility that independent national monetary policy provides, or a pure deadweight loss imposed by the abandonment of a common standard, is again a matter on which reasonable economists disagree, and this essay does not propose to settle it.

CHAPTER SEVEN

Money and the Shape of Time

Beneath the technical questions of stock-to-flow ratios and exchange-rate mechanics sits a psychological claim that the Austrian tradition treats as central: the soundness of a society's money shapes how much its members discount the future, a concept economists call time preference. It is worth taking this claim seriously and, in the same breath, being honest about how far past the available evidence it is sometimes pushed.

The uncontroversial part

The basic mechanism is not particularly controversial. If a currency reliably loses purchasing power year after year, holding cash balances for the long term becomes a losing proposition, and rational actors respond at the margin by consuming sooner rather than later, or by directing savings toward assets that are expected to outpace inflation, which frequently means assets carrying more risk than a saver might otherwise choose. If a currency instead holds its value, or gently appreciates, patience is rewarded rather than punished, and the incentive to defer consumption in favour of saving and investment strengthens. This is standard economics, traceable through Eugen von Böhm-Bawerk's work on capital and interest, and it explains, among other things, why persistently negative real interest rates during the 2010s coincided with a well-documented reach for yield across pension funds and retail investors alike, as savers hunted for returns that would simply preserve, rather than erode, the value of what they had already saved.

The more contentious part

Where the argument becomes considerably shakier is in its extension from individual financial decisions to broad claims about culture, family structure, and artistic achievement. Saifedean Ammous, whose popularisation of this tradition has been influential, argues at some length that the flourishing of the arts and sciences in the late nineteenth-century gold standard era, and their supposed decline since, can be traced to the soundness of the money underwriting each period. It is an evocative argument, and it is also the kind of sweeping historical claim that professional historians and economists tend to treat with considerable caution, because the late nineteenth century was also the era of Western imperial expansion, mass industrialisation, and a scientific revolution with causes that most historians would locate in institutions, demographics, and technology diffusion rather than in the metallic backing of contemporary banknotes. Correlation across a single historical period, however striking, is thin evidence for a causal claim about something as multiply determined as artistic output or family stability, and readers should hold this part of the tradition much more loosely than the arithmetic of stock-to-flow ratios discussed earlier, which rests on considerably firmer ground.

None of this means time preference is an unimportant concept, only that its explanatory reach should be matched carefully to the strength of the evidence behind each claim. The link between currency debasement and reduced willingness to save is well supported. The link between currency debasement and, say, the decline of a particular school of painting is asserted far more often than it is demonstrated.

CHAPTER EIGHT

The Business-Cycle Question

No part of the Austrian tradition generates more disagreement among professional economists than its theory of the business cycle, so it is worth stating plainly, up front, that what follows is a description of a heterodox position, not the consensus view taught in most university economics departments.

The Austrian account, briefly

The Austrian business cycle theory, developed principally by Ludwig von Mises and elaborated by Friedrich Hayek, holds that recessions are not random shocks to be smoothed over with fresh stimulus, but the inevitable unwinding of an artificial boom created by interest rates held below their natural market level. When a central bank expands credit and pushes interest rates down without a corresponding increase in genuine savings, the theory goes, entrepreneurs are misled into believing more real capital is available for long-term projects than actually exists. They begin building things that require more resources to finish than society has actually set aside through deferred consumption — Hayek's own metaphor was of a builder who starts more houses than he has bricks to complete. Eventually the shortage becomes visible, projects are abandoned mid-construction, and the resulting bust is not the disease but the necessarily painful correction of a boom that never should have happened in that shape.

Why most economists remain unconvinced

The mainstream case against this theory, made forcefully by monetarists and Keynesians alike despite their other disagreements, runs roughly as follows. Empirically, recessions frequently arrive without any preceding episode of unusually loose central bank credit, which the pure Austrian account struggles to explain. The theory also has an awkward relationship with unemployment: if a recession is simply the necessary liquidation of malinvested capital, it is not obvious why that liquidation should require large numbers of otherwise employable workers to sit idle for extended periods, rather than being reabsorbed quickly into other, more soundly financed lines of production. And attempts to test the theory rigorously against historical data, going back to work by economists sympathetic to neither Austrian nor strict monetarist priors, have generally found its predictions difficult to distinguish from ordinary demand-side accounts of the business cycle.

None of this makes the Austrian account worthless as a lens. Its central insight — that artificially cheap credit can push capital into projects that would not have been undertaken at honest interest rates, and that the resulting misallocation has to be worked through somehow — is a genuine contribution, and something like it shows up, in softened form, in mainstream accounts of asset bubbles and the 2008 financial crisis specifically. But the strong version of the claim, that essentially all recessions trace to central bank interest-rate manipulation and that a commodity-backed money would eliminate the business cycle rather than merely changing its shape, remains a minority position within the economics profession, and this essay is not in a position to adjudicate a decades-old dispute that has resisted resolution by economists considerably better equipped to resolve it.

WHY THIS MATTERS FOR WHAT FOLLOWS

The Austrian account of the business cycle is frequently cited as a reason to prefer a monetary system, such as Bitcoin's, whose supply cannot be discretionarily expanded by any central authority. Whether one finds that argument persuasive should not depend on treating the underlying business-cycle theory as more settled than it actually is. There are other, more straightforwardly uncontested reasons — discussed in the next chapter — to find a fixed-supply digital asset interesting, and they do not require buying the whole Austrian package to take seriously.

CHAPTER NINE

Digital Scarcity

Every monetary good examined so far, however hard to produce, has ultimately been vulnerable to some change in the cost of producing more of it: a new tool, a new trade route, a new source of supply. Gold's chemistry made it the hardest of the physical candidates, but chemistry is a fact about atoms, not an unbreachable law, and asteroid mining, however distant a prospect, is at least theoretically capable of one day disrupting gold's stock-to-flow ratio in a way that centuries of terrestrial mining never could. Bitcoin's proponents argue that its 2009 invention solved this problem in a genuinely novel way, and it is worth understanding the mechanism carefully before deciding how much weight to put on the claim.

What proof of work actually does

Bitcoin's ledger of transactions is maintained not by a trusted institution but by a competitive network of participants, called miners, who expend real computing power and electricity solving a deliberately difficult mathematical puzzle for the right to add the next batch of transactions to the shared record. This is expensive by design: the difficulty of the puzzle adjusts automatically, roughly every two weeks, so that a new batch of transactions is added on average every ten minutes regardless of how much computing power the network has attracted. The purpose of this expense is to make dishonesty costly. Falsifying the ledger would require redoing this energy-intensive work faster than the rest of the honest network combined, an undertaking that becomes more expensive, not less, the longer the honest chain grows, since an attacker must out-race an ever-lengthening history rather than a fixed target.

The reward for successfully adding a batch of transactions is a fixed quantity of newly issued bitcoin, and this issuance rate is cut in half on a fixed schedule, roughly every four years, until the total supply asymptotically approaches a hard cap of 21 million coins, expected to be substantially exhausted by the mid-2030s and fully so around the year 2140. Crucially, this schedule does not respond to price. When gold's price rises sharply, more mining becomes profitable and, given enough time, more gold gets mined. When bitcoin's price rises sharply, the reward per block stays exactly where the code says it should; only the computing power competing for that fixed reward increases, which makes the network more secure but does not create a single additional coin. This is, on the evidence available so far, a genuinely new phenomenon: a monetary good whose issuance schedule is arguably more resistant to influence by rising demand than even gold's has ever been.

What this costs, and what remains genuinely unresolved

None of this comes free, and intellectual honesty requires sitting with the costs rather than skating past them. The security model depends on genuinely large expenditures of electricity, a fact that has made bitcoin mining a recurring subject of environmental scrutiny, and defenders and critics of the network continue to dispute, sometimes heatedly, how much of that electricity draws on stranded or otherwise wasted energy sources versus displacing capacity that could serve other purposes. Bitcoin's price has also been extremely volatile across its history, which sits awkwardly with any claim that it currently functions well as a unit of account or a stable store of value over short horizons, whatever its long-run trajectory has looked like to early holders; a genuinely sound money, on the definitions this essay opened with, ought to hold its value reasonably steadily, and bitcoin, as of this writing, plainly does not, even if its defenders argue this volatility should be expected to diminish as the market matures and the currency captures a larger share of global monetary demand.

There are further open questions that this essay will not pretend to close. The network's transaction throughput is deliberately limited, which has pushed much day-to-day activity onto secondary layers, such as the Lightning Network, that trade some of the base layer's trust-minimisation for speed and lower cost — a reasonable engineering compromise, but one that reintroduces an element of counterparty trust that the base protocol was specifically designed to avoid. Custody remains a genuinely hard problem for ordinary users: a private key is unforgiving of loss or theft in a way that a bank account, whatever its own vulnerabilities, generally is not. And the regulatory environment surrounding the asset continues to shift across jurisdictions in ways that are difficult to forecast with any confidence.

What can be said with reasonable confidence is narrower than the boldest claims made on bitcoin's behalf, but still notable: it is the first digital good whose scarcity does not depend on anyone's promise to keep it scarce, verifiable by anyone running the software rather than trusted on the word of an issuer, and its issuance schedule has, for a decade and a half, behaved exactly as its original design specified, through several attempts by disagreeing factions within its own community to change it. Whether that is sufficient to make it a durable store of value over a horizon of decades, rather than years, is a question this essay is not old enough to answer, and neither, yet, is bitcoin.

CHAPTER TEN

Open Questions

It would be tidy to end with a single verdict — gold good, fiat bad, bitcoin the inevitable heir — and it would also misrepresent the state of the argument this essay has been walking through. A more honest closing runs along different lines.

The saleability framework inherited from Menger, and the stock-to-flow arithmetic built on top of it, are genuinely useful tools for understanding why certain goods have functioned well as money across history and why others have not. That much of the argument rests on solid ground, is testable against the historical record, and does not require accepting any particular political conclusion to find persuasive. The claim that unsound money erodes a society's willingness to save is likewise reasonably well supported, both theoretically and in the observed behaviour of savers under high-inflation regimes.

Beyond that point, the tradition this essay has been exploring makes considerably stronger claims than the evidence comfortably supports: that the business cycle is fully explained by central bank interest-rate manipulation, that a commodity-backed or algorithmically fixed money would eliminate recessions rather than reshape them, and that the cultural and artistic character of an era can be read off the metallic content of its coinage. These are live debates within economics and economic history, not settled questions, and a reader who encounters them presented with total confidence, in either direction, should treat that confidence itself as a signal to look more closely rather than less.

What does seem clear, and what this essay would defend without much hedging, is that the question of who controls the supply of a society's money, and by what rules, is not a narrow technical matter best left to specialists. It has been, across every episode surveyed here, from Yap to Weimar to the present debate over digital scarcity, one of the more consequential design choices any society makes, and one that repays a bit of monetary history before forming a view.

APPENDIX

For Further Reading

This essay is a synthesis and, in places, a friendly cross-examination of a specific intellectual tradition. Readers who want the primary sources, including the arguments this essay pushes back against, should go to them directly rather than take this summary's word for any of it.

— **Carl Menger**, "On the Origin of Money" (1892) — the founding text on saleability and the spontaneous emergence of money.

— **Ludwig von Mises**, *The Theory of Money and Credit* (1912) and *Human Action* (1949) — the fullest statement of the Austrian monetary and business-cycle tradition.

— **Friedrich Hayek**, *Monetary Nationalism and International Stability* (1937) and "The Use of Knowledge in Society" (1945).

— **R. A. Radford**, "The Economic Organisation of a P.O.W. Camp," *Economica* (1945) — the cigarette-currency study referenced in Chapter Four.

— **Milton Friedman & Anna Schwartz**, *A Monetary History of the United States, 1867–1960* (1963) — the monetarist counterpoint to the Austrian account of the Great Depression.

— **Barry Eichengreen**, *Golden Fetters: The Gold Standard and the Great Depression, 1919–1939* (1992) — a mainstream economic history of the interwar gold standard's collapse.

— **Saifedean Ammous**, *The Bitcoin Standard* (2018) — the contemporary popularisation this essay is in dialogue with throughout.

— **Satoshi Nakamoto**, "Bitcoin: A Peer-to-Peer Electronic Cash System" (2008) — the original technical proposal, nine pages, and worth reading in full.

Prepared for The Ratty2Tails Papers, Monetary History Series. This essay reflects the editor's synthesis of the sources above and does not reproduce the text of any of them; where a specific claim, figure, or framing belongs to a named author, an effort has been made to say so in the body of the essay.